

# Half-Collinear Graviton Vertices as Weighted Root-Cone Enumerators

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## Abstract

The general half-collinear single-minus graviton amplitude recently obtained by Guevara, Lupsasca, Skinner, Strominger, and Weil is expressed through retarded and advanced multi-point vertices whose Cayley-tree weights contain nonlocal step functions. We show that each retarded vertex is equivalently a weighted enumerator of directed spanning-tree root cones containing a kinematically defined netflow vector; the advanced vertex evaluates the opposite cones. Consequently, the kinematic walls are contained among the hyperplanes of the resonance arrangement. This resolves the published matrix-tree obstruction in a class of chambers: the global cut conditions become root-cone membership, and whenever the feasible family is arborescent the ordinary directed matrix-tree theorem applies. After ordering the projective slopes of the block spinors, we characterize which labeled resonance chambers are realizable by generic real two-spinors and show that their number is  $(m-1)R_{m-1}/m$  for an  $m$ -block vertex, where  $R_{m-1}$  is the number of rank- $(m-1)$  resonance chambers. An exhaustive classification gives 24 realizable chambers at four blocks and 296 at five blocks. Six nontrivial four-block chambers and 57 nontrivial five-block chambers reduce to directed arborescence enumerators and therefore admit matrix-tree determinant formulas. As an explicit application, we derive a compact five-graviton formula in a chamber outside the previously studied decay region. All low-rank classifications and the five-point identity are verified with exact arithmetic.

## 1 Introduction

Reference [1] established that single-minus tree-level graviton amplitudes, often presumed to vanish, are nonzero on half-collinear configurations in Klein signature or at complexified momenta. The authors derived an on-shell Berends–Giele recursion and an explicit solution in terms of unordered Feynman trees [3]. In a special decay chamber, the answer collapses to a product of soft factors after an application of the directed matrix-tree theorem, in parallel with determinant formulas familiar from graviton amplitudes [4]. Simplification in general kinematics was left open; the source paper also notes that the global tree dependence prevents a direct matrix-tree treatment of its general formula.

The obstruction is easy to state. A Cayley tree contributes only when every edge passes a sign test involving the total momentum on both sides of the cut created by deleting that edge. These tests depend on the global tree, so the usual edge-local matrix-tree theorem does not immediately apply.

Our main observation is that the global cut test has a standard interpretation: it asks whether a prescribed netflow is generated by strictly positive flows on the directed tree. This identifies the graviton vertex with a weighted root-cone enumerator and imports the chamber combinatorics of the resonance arrangement [2]. The reformulation produces three concrete gains:

1. a geometric containment and chamber classification of the kinematic walls;
2. a count and constructive parameterization of real-spinor-realizable low-rank chambers; and
3. new determinant and compact amplitude formulas in chambers beyond the decay region.

The appearance of this arrangement around objects called “retarded” is not without precedent. Resonance chambers count generalized retarded functions in axiomatic and thermal quantum field theory [5, 6], and the adjoint braid arrangement supports a modern combinatorial formulation of Steinmann relations and their factorization properties [7, 8]. Those works do not identify the specific half-collinear graviton vertex studied here with a weighted root-cone enumerator. They nevertheless show that the general link between retarded objects and resonance chambers is classical; the new claims below are the explicit flow-cone equivalence, the realizability theorem, the arborescent classification, and the non-decay five-point formula.

This manuscript is a working research draft. Its claims and computations have passed an adversarial independent machine review, but they have not yet been checked by an unaffiliated human expert in scattering amplitudes or resonance arrangements.

## 2 The half-collinear multipoint vertex

Let  $q_1, \dots, q_m \in \mathbb{R}^2$  denote the real two-spinors entering a block vertex, let

$$Q = \sum_{i=1}^m q_i, \quad [A, B] = \left[ \sum_{i \in A} q_i, \sum_{j \in B} q_j \right], \quad (1)$$

and write  $[ij] = [q_i, q_j]$  for the antisymmetric determinant. We assume generic data: no pair bracket and no cut bracket appearing below vanishes.

For a Cayley tree  $T$  on  $[m] = \{1, \dots, m\}$ , deleting an edge  $e = \{u, v\}$  produces a cut  $A_e \sqcup B_e = [m]$ . Choosing  $u \in A_e$ , the retarded vertex of Ref. [1] is

$$V(q_1, \dots, q_m) = \sum_T \prod_{e=\{u,v\} \in E(T)} |[uv]| \Theta \left( -\frac{[A_e, B_e]}{[uv]} \right). \quad (2)$$

The choice is immaterial: simultaneously exchanging  $(u, A_e) \leftrightarrow (v, B_e)$  reverses both numerator and denominator in the step-function argument. This agrees with the source convention, which fixes  $u$  to be the smaller endpoint label. The advanced vertex  $\bar{V}$  is obtained by reversing the sign in every step-function argument, and the LSZ kernel is  $\hat{T} = V - \bar{V}$ .

The apparent nonlocality lies entirely in  $[A_e, B_e]$ , which depends on the whole component produced by cutting  $e$ .

## 3 Root-cone representation

Define the projected vector  $x \in \mathbb{R}^m$  by

$$x_i = [q_i, Q]. \quad (3)$$

Antisymmetry gives  $\sum_i x_i = [Q, Q] = 0$ . Direct every edge of the complete graph by

$$i \longrightarrow j \iff [ij] < 0. \quad (4)$$

Every undirected Cayley tree inherits this orientation.

For a directed graph  $G$  on  $[m]$ , its open root cone is

$$C^\circ(G) = \left\{ \sum_{i \rightarrow j \in E(G)} f_{ij}(e_i - e_j) : f_{ij} > 0 \right\}. \quad (5)$$

Thus  $x \in C^\circ(G)$  precisely when  $x$  is the divergence vector of a strictly positive flow on  $G$ .

**Theorem 3.1** (Flow-cone form of the graviton vertex). *Away from chamber walls,*

$$V(q_1, \dots, q_m) = \sum_{\substack{T \text{ spanning tree} \\ x \in C^\circ(T)}} \prod_{\{i,j\} \in E(T)} |[ij]|. \quad (6)$$

*If every tree is given the retarded orientation (4), then*

$$\bar{V}(q_1, \dots, q_m) = \sum_{\substack{T \text{ spanning tree} \\ -x \in C^\circ(T)}} \prod_{\{i,j\} \in E(T)} |[ij]|. \quad (7)$$

*Proof.* For the cut induced by deleting  $e$ ,

$$[A_e, B_e] = [q_{A_e}, Q - q_{A_e}] = [q_{A_e}, Q] = \sum_{i \in A_e} x_i. \quad (8)$$

A divergence vector on a directed tree fixes every edge flow uniquely. Its flow through  $e$  has magnitude  $|\sum_{i \in A_e} x_i|$ . If  $[uv] > 0$ , the retarded orientation is  $v \rightarrow u$ , the edge enters  $A_e$ , and positivity requires  $\sum_{i \in A_e} x_i < 0$ . If  $[uv] < 0$ , the edge leaves  $A_e$  and positivity requires the opposite inequality. Both cases are exactly

$$\Theta\left(-\frac{[A_e, B_e]}{[uv]}\right) = 1. \quad (9)$$

The tree survives all step functions if and only if  $x \in C^\circ(T)$ , and its remaining weight is the product in (6). Reversing every step-function sign reverses every tree orientation, whose root cone is  $-C^\circ(T)$ .  $\square$

**Corollary 3.2** (Resonance walls). *For fixed pair-bracket signs, every vertex wall is a resonance hyperplane*

$$\sum_{i \in A} x_i = 0, \quad \emptyset \neq A \subsetneq [m]. \quad (10)$$

*Consequently,  $V$  and  $\bar{V}$  are weighted tree enumerators that are constant in tree support on every resonance chamber.*

## 4 Which resonance chambers are realizable?

The resonance arrangement contains abstract sign chambers that need not occur for real spinors with a fixed projective ordering. The realizability restriction is simple.

Choose an  $SL(2, \mathbb{R})$  frame with  $Q = (1, 0)$ , assume every  $x_i \neq 0$ , and write

$$q_i = (x_i r_i, -x_i). \quad (11)$$

After relabeling, take  $r_1 < \dots < r_m$ . Momentum conservation gives

$$\sum_i x_i = 0, \quad \sum_i x_i r_i = 1, \quad (12)$$

while

$$[ij] = x_i x_j (r_j - r_i). \quad (13)$$

Thus the pair-bracket tournament is fixed by the signs of  $x_i$  once the slopes are ordered.

**Theorem 4.1** (Realizable chamber criterion). *Let  $X_k = \sum_{i=1}^k x_i$ . A generic resonance chamber—so in particular all  $x_i$  and all proper subset sums, including the prefix sums, are nonzero—has a realization with  $r_1 < \dots < r_m$  and (12) if and only if*

$$X_k < 0 \quad \text{for at least one } k \in \{1, \dots, m-1\}. \quad (14)$$

*Equivalently, the only excluded chambers are those in the positive root cone aligned with the slope order.*

*Proof.* Summation by parts yields

$$\sum_{i=1}^m x_i r_i = - \sum_{k=1}^{m-1} X_k (r_{k+1} - r_k). \quad (15)$$

Every slope gap is positive. Hence the right-hand side can equal 1 only if some  $X_k < 0$ . Conversely, if one prefix is negative, choose its slope gap sufficiently large relative to all other positive gaps so that the right-hand side is positive, and rescale all gaps to make it equal to 1.  $\square$

Let  $R_{m-1}$  be the number of rank- $(m-1)$  resonance chambers. The cyclic decomposition in Ref. [2] shows that exactly  $R_{m-1}/m$  lie in the aligned positive root cone. Theorem 4.1 therefore gives the number of labeled, slope-ordered realizable chambers,

$$R_{m-1}^{\text{real}} = \frac{m-1}{m} R_{m-1}. \quad (16)$$

*Remark 4.2.* Here “realizable” is a kinematic statement about generic real two-spinors, not a claim that every counted chamber lies in a fixed scattering channel. The decay region of Ref. [1] additionally imposes energy signs and therefore selects a smaller in/out configuration.

## 5 Determinantal chambers and low-rank classification

For a fixed realizable chamber  $\mathcal{C}$ , let  $\mathcal{F}_{\mathcal{C}}$  be the family of retarded-oriented spanning trees whose open root cones contain  $x$ . Theorem 3.1 is always a weighted sum over this family. Some chambers restore an edge-local description.

**Definition 5.1.** A chamber is *arborescent* if there are a directed graph  $D_{\mathcal{C}}$  and a root  $\rho$  such that  $\mathcal{F}_{\mathcal{C}}$  is exactly the set of all directed spanning trees of  $D_{\mathcal{C}}$  oriented toward  $\rho$ .

Give every arc  $i \rightarrow j$  weight  $w_{ij} = |[ij]|$  and define the outgoing Laplacian

$$L_{ij} = \begin{cases} -w_{ij}, & i \rightarrow j, \ i \neq j, \\ \sum_{i \rightarrow k} w_{ik}, & i = j, \\ 0, & \text{otherwise.} \end{cases} \quad (17)$$

**Proposition 5.2** (Determinantal chamber formula). *In an arborescent chamber,*

$$V(q_1, \dots, q_m) = \det L^{(\rho)}, \quad (18)$$

where  $L^{(\rho)}$  deletes row and column  $\rho$ .

*Proof.* This is the directed matrix-tree theorem combined with Theorem 3.1 and the definition of an arborescent chamber.  $\square$

We exhaustively enumerated the known 32 rank-three and 370 rank-four resonance chambers. The realizability criterion leaves 24 four-block and 296 five-block chambers. For each realizable chamber we constructed an exact integer representative, enumerated all  $m^{m-2}$  Cayley trees, tested positive-flow feasibility, and checked whether the resulting family was the complete in-arborescence family of its union digraph. Table 1 summarizes the result.

|                                   | $m = 4$ | $m = 5$ |
|-----------------------------------|---------|---------|
| All resonance chambers            | 32      | 370     |
| Realizable ordered-slope chambers | 24      | 296     |
| Distinct feasible-tree families   | 9       | 55      |
| Chambers with $V = 0$             | 6       | 38      |
| Nonempty arborescent chambers     | 10      | 57      |
| Nontrivial arborescent chambers   | 6       | 57      |
| Largest feasible-tree family      | 5       | 28      |

Table 1: Exhaustive low-rank chamber classification. Distinct feasible-tree families and every subsequent row are counted only over realizable labeled chambers. “Nontrivial” means more than one contributing spanning tree.

All 67 reported determinant identities were independently evaluated as exact Laplacian cofactors and compared with the original step-function sums. At five blocks every nonempty arborescent family is already nontrivial: the observed feasible-family sizes jump directly from zero to two.

## 6 A compact non-decay five-graviton amplitude

We now lift one arborescent four-block chamber through the five-point recursion. Let  $q_1, \dots, q_4$  be the positive-helicity block spinors,  $Q = q_1 + \dots + q_4$ , and  $q_5 = -Q$ . Define

$$x_i = [q_i, Q] = -[i5], \quad y_i^S = [q_i, q_S] \quad (i \in S). \quad (19)$$

Consider the open sign chamber  $\mathcal{C}_\star$  specified in Table 2. The pair signs fix the retarded tournament; the  $x$  signs fix the four-block resonance chamber; and the  $y$  signs fix the four three-block subvertices entering the recursion.

Writing  $w_{ij} = |[ij]|$ , direct root-cone inspection gives

$$\mathcal{F}(V_{1234}) = \{\{14, 23, 24\}, \{14, 24, 34\}\}, \quad (20)$$

$$\mathcal{F}(\bar{V}_{1234}) = \{\{12, 13, 14\}, \{13, 14, 23\}\}. \quad (21)$$

Both are two-term arborescence families, so

$$V_{1234} = w_{14}w_{24}(w_{23} + w_{34}), \quad (22)$$

$$\bar{V}_{1234} = w_{13}w_{14}(w_{12} + w_{23}). \quad (23)$$

| Data              | Ordered quantities                                 | Signs                |
|-------------------|----------------------------------------------------|----------------------|
| Projected singles | $(x_1, x_2, x_3, x_4)$                             | $(-, +, +, -)$       |
| Projected pairs   | $(x_{12}, x_{13}, x_{14}, x_{23}, x_{24}, x_{34})$ | $(-, -, -, +, +, +)$ |
| Pair brackets     | $([12], [13], [14], [23], [24], [34])$             | $(-, -, +, +, -, -)$ |
| $S = 123$         | $(y_1^{123}, y_2^{123}, y_3^{123})$                | $(-, +, +)$          |
| $S = 124$         | $(y_1^{124}, y_2^{124}, y_4^{124})$                | $(+, +, -)$          |
| $S = 134$         | $(y_1^{134}, y_3^{134}, y_4^{134})$                | $(-, +, -)$          |
| $S = 234$         | $(y_2^{234}, y_3^{234}, y_4^{234})$                | $(+, -, +)$          |

Table 2: Definition of the open five-point chamber  $\mathcal{C}_*$ . Here  $x_{ij} = x_i + x_j$ .

The triple sign vectors similarly imply

$$V_{234} = V_{123} = 0, \quad V_{134} = w_{14}w_{34}, \quad V_{124} = w_{12}w_{24}. \quad (24)$$

The published five-point recursion is

$$M_{12345} = -\hat{T}_{1234} - w_{15}V_{234} - w_{25}V_{134} - w_{35}V_{124} - w_{45}V_{123}, \quad (25)$$

with  $\hat{T} = V - \bar{V}$ . To obtain this form directly from Eq. (33) of Ref. [1], use the preamplitude base cases  $\bar{M}_i = 1$ ,  $\bar{M}_{ij} = 0$ , and  $\bar{M}_{ijk} = -V_{ijk}$ . Only the four-singleton partition and the four partitions consisting of one singleton plus one triple then survive. Substitution proves the following.

**Corollary 6.1** (Five-graviton formula outside the decay chamber). *Throughout  $\mathcal{C}_*$ ,*

$$\begin{aligned} M_{12345} = & w_{13}w_{14}(w_{12} + w_{23}) - w_{14}w_{24}(w_{23} + w_{34}) \\ & - w_{25}w_{14}w_{34} - w_{35}w_{12}w_{24}. \end{aligned} \quad (26)$$

*This is a genuine non-decay formula: both  $V_{1234}$  and  $\bar{V}_{1234}$  are nonzero in  $\mathcal{C}_*$ .*

The proof is finite and analytic: Table 2 determines every step function in the 16 four-vertex Cayley trees and the three trees of each triple subvertex. Equation (26) is therefore constant in form on the entire open chamber, not merely on sampled kinematics.

## 7 Exact computational verification

The accompanying Python implementation uses integer two-spinors and exact arithmetic. No floating tolerances enter any sign, determinant, or amplitude comparison. We use  $[a, b] = a_0b_1 - a_1b_0$ , globally opposite to the component ordering used for brackets in Ref. [1]; an  $n$ -point homogeneous formula therefore acquires the expected factor  $(-1)^{n-2}$  when transcribed directly. The following checks currently pass:

- direct assertions of  $M_{123} = |[12]|$  and  $M_{1234} = \frac{1}{2}(|[12]||[34]| + |[13]||[24]| + |[14]||[23]|)$  on 100 exact samples;
- agreement with the published decay-region product formula at four and five points on 100 exact samples, including the bracket-convention sign;
- permutation invariance through five points on randomized samples;

- agreement of the step-function and positive-flow definitions of both  $V$  and  $\overline{V}$  on 500 randomized samples through valency six;
- complete recovery of the known 32 and 370 resonance chambers from bounded integer representatives;
- exact matrix-tree cofactor checks in all 10 four-block and 57 five-block nonempty arborescent chambers; and
- 22,102 exact integer tests of (26) across  $\mathcal{C}_*$ , obtained by bounded grid enumeration rather than random sampling.

Two additional referee programs were written from scratch without importing the bundled implementation. They independently reconstruct the published recursion, reproduce the decay formula, verify the flow-cone equivalence on 300 exact samples, recover Table 1, and derive the families in (22)–(24) from the chamber signs alone. These machine checks are evidence against implementation coupling; they do not replace expert proof review.

Two tempting stronger conjectures fail. First, the full amplitude is not the sum over local cubic-tree vertex weights suggested by the three- and four-point answers; it fails at generic five-point kinematics. Second, the generic five-point answer is not a permutation-symmetric homogeneous cubic polynomial in the ten variables  $[[ij]]$ : the seven graph-monomial orbit sums have rank seven on exact data, while adjoining the amplitude raises the rank to eight. These failures support the conclusion that resonance cut data is essential outside special chambers.

## 8 Discussion

The root-cone formulation does not produce a single determinant valid in every chamber. It instead identifies the precise combinatorial object that replaces one: a weighted collection of tree cones containing a netflow vector. This is useful because resonance chambers, root-cone intersections, compatible tree collections, and Kostant chambers already have a substantial mathematical theory [2].

Accordingly, the result should be read as a precise partial resolution of the matrix-tree obstruction stated in Ref. [1], not as a closed form for the unrestricted amplitude. The global cut tests are converted into root-cone membership for every chamber; the ordinary matrix-tree theorem then applies exactly when the feasible family is arborescent, a condition classified here through five blocks.

Several next questions are concrete.

1. Characterize arborescent chambers without exhaustive tree enumeration.
2. Determine whether non-arborescent families admit decompositions into a small signed sum of Laplacian cofactors.
3. Lift the 57 five-block determinant identities to compact six-graviton formulas, tracking the required lower-valency subchambers.
4. Relate chamber adjacency to wall-crossing identities of the amplitude.
5. Investigate whether the root-cone language clarifies the extension of the  $\mathcal{L}w_{1+\infty}$  recursion outside the decay region.

The most important immediate task is independent verification by experts in scattering amplitudes and resonance arrangements. In particular, absence of the present connection from the cited graviton paper does not by itself establish novelty across the full literature.

## Reproducibility statement

The release is pinned to Python 3.12.3 and is accompanied by three exact-arithmetic programs: one implementing and anchoring the published recursion and flow-cone equivalence, one enumerating and classifying low-rank chambers, and one exhaustively testing the five-point identity. It also contains the two independently written referee programs, machine-readable certificates for every realizable four- and five-block chamber, cryptographic checksums, and continuous-integration tests.

## Author and AI contribution statement

James Kehoe selected and directed the research problem, made the decision to pursue and disseminate the result, and commissioned adversarial review. OpenAI Codex generated the central root-cone reformulation, proofs, search and verification code, and initial manuscript under that direction. An independent large-language-model review then re-derived the main claims using fresh code; its report and programs are included in the repository. No AI system is listed as an author. The named human author accepts responsibility for the released manuscript and will update or withdraw claims if subsequent expert review finds an error.

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